

Applications



1

a $f(t) = 250 + 100 \sin 2\pi t$

c 36 times

e $t = \frac{7}{12}$

b $g(t) = 350 - 100 \sin 4\pi t$

d $\frac{1}{8}$ years

2

a $f(t) = 150 + 40 \sin 2\pi t$

c 27 times

b $g(t) = 100 - 40 \sin 3\pi t$

d $t = 1.70$

3

a Sine, the graph's initial value is at the midline.

b $I = \sin\left(\frac{\pi t}{2}\right)$

c 210 times

4

a 6 seconds

b 2500 mL

c 200 mL

d 60° per second

e $c = \frac{1}{2}$

5



a

i No

ii No

iii No

iv Yes

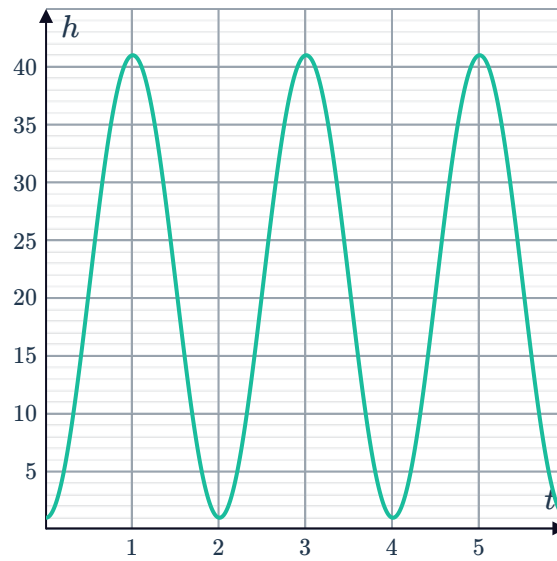
b

i 2 minutes

ii 1 m

iii 41 m

c



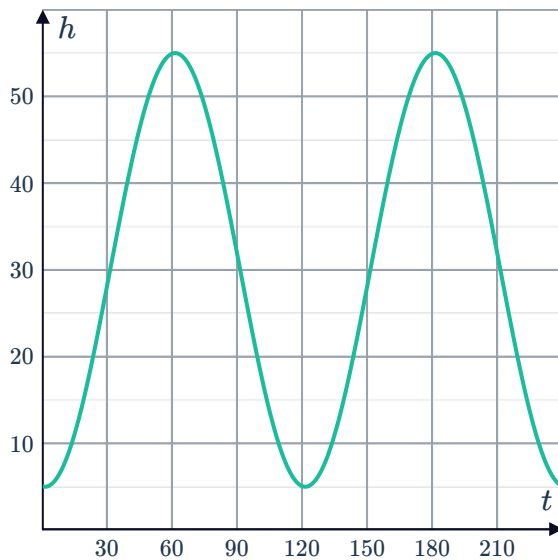
d $h = -20 \cos(\pi t) + 21$

e $t = \frac{1}{2}$ minutes

f The height of the passengers above or below the centre of the wheel at time t .

6

a



b 55 m

c 5 m

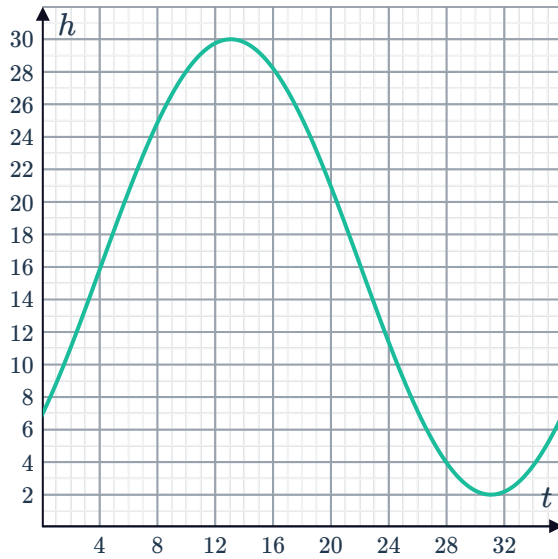
d 36.47 m

e 120 seconds

7



a



b 7.00 m

c $t = 34.82$ seconds

d 3

e 3.88 m

8

a

i Yes

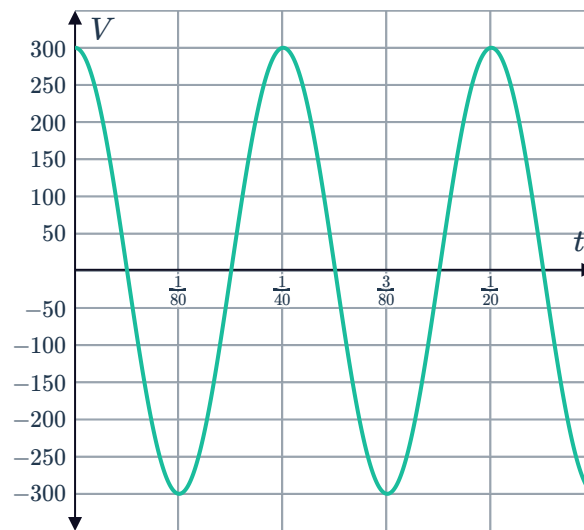
ii No

iii No

iv No

b $V = 300 \cos(80\pi t)$

c



d It would decrease the period of the function and compress the graph horizontally.

9



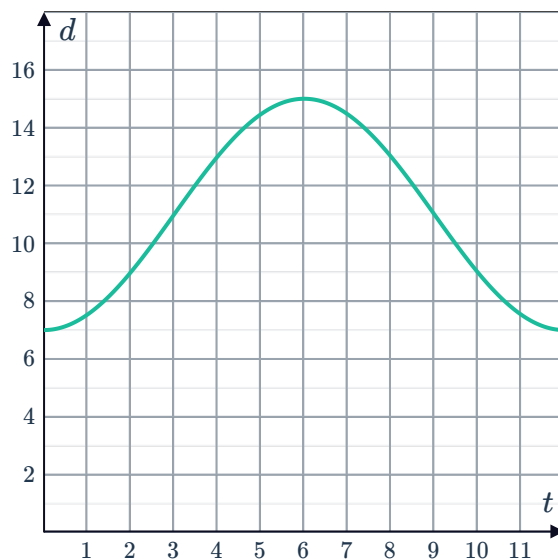
a

Time (t , hours)	0	3	6	9	12
Tide level (d , metres)	7	11	15	11	7

b 4 m

c 12 hours

d



e

i Yes

ii No

iii No

iv No

f $d = -4 \cos \frac{\pi}{6} t + 11$

g 2.66 hours

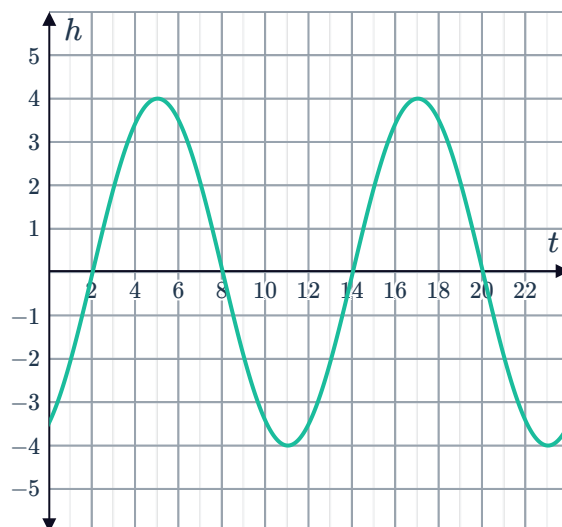
10

a -2 m

c 6 hours

b 5:00 am

d



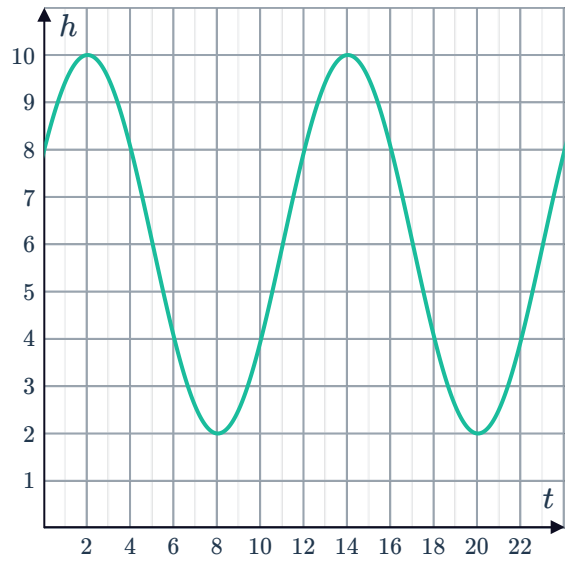
e 0 m

f 0.14 m

a $h(t) = 6 + 4 \cos\left(\frac{\pi}{6}(t - 2)\right)$



b



c 2:00 am

d 8:00 am

12

a

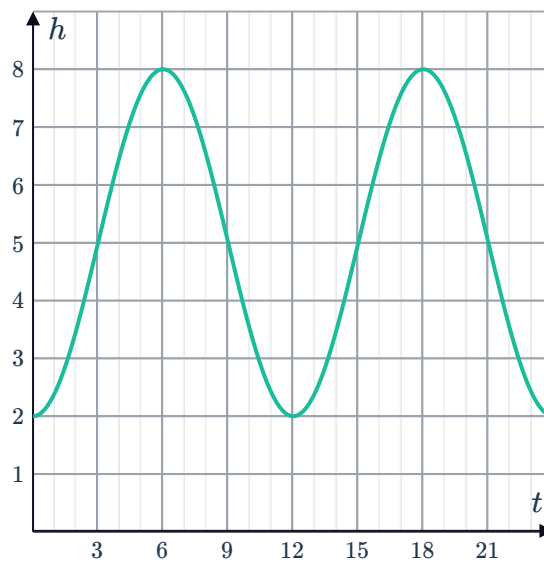
i $d = 5$ m

ii $a = 3$ m

iii $b = 30^\circ/\text{h}$

iv $c = 90^\circ$

b



c 6 and 18 hours after noon

13

a

i $a = 40$ cm

ii $d = 10$ cm

iii $c = \frac{1}{4}$

b $t = \frac{1}{3}$ seconds

14

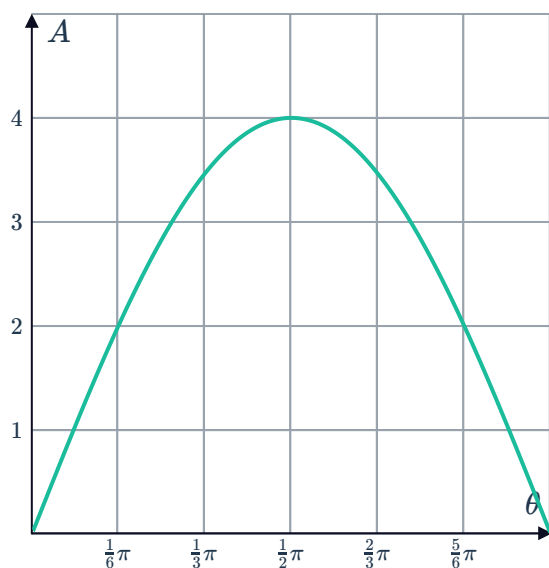
a $A = 4 \sin \theta$



b $0 < \theta < \pi$

c

d $\frac{\pi}{6}$



15

a 5 cm to the right of the centre

b To the left

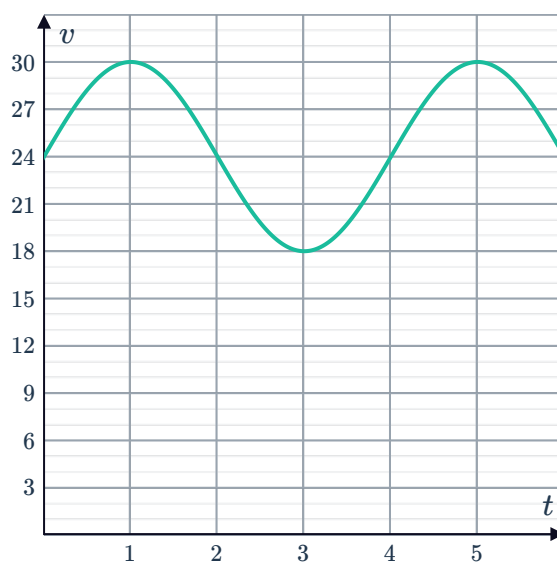
c 1 second

d 4 seconds

e $x = 5 \cos \left(\frac{\pi t}{2} \right)$

16

a



b 30 units

c The number of seconds that have passed since the beginning of the section.

d 24 units

17

a 1.44 seconds

b 12.46 hours

c 110 days